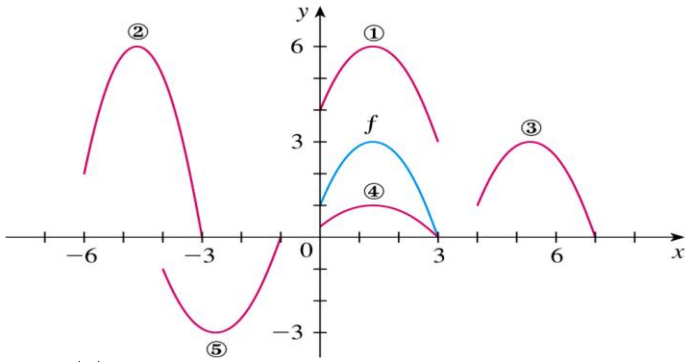
UCCM1153 Introduction to Calculus and Applications

T1: Functions and Models

1. Find the domain and range and sketch the graph of the function $h(x)=\sqrt{4-x^2}$.

2. Find the domain and sketch the graph of the function $F(x)=|2x+1|$.

3.




The graph of $y=f(x)$ is given. Match each equation with its graph and give reasons for your choices.

(a) $y=f(x-4)$

(b) $y=f(x)+3$

(c) $y=\frac{1}{3}f(x)$

(d) $y=-f(x+4)$

(e) $y=2f(x+6)$

4.


The graph of $f$ is given. Use it to graph the following functions.

(a) $y=f(2x)$

(b) $y=f\left(\frac{1}{2}x\right)$

(c) $y=f(-x)$

(d) $y=-f(-x)$

5. Graph the function by hand, not by plotting points, but by starting with the graph of one of the standard functions, and then applying the appropriate transformations.

(a) $y=4\sin 3x$

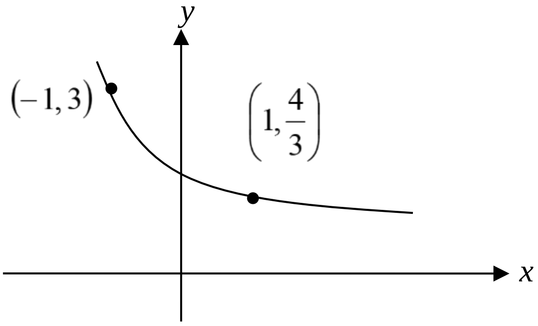
(b) $y=1+\sqrt[3]{x-1}$

6. Find $f\circ g\circ h$.

$f(x)=\tan x,\ g(x)=\dfrac{x}{x-1},\ h(x)=\sqrt[3]{x}.$

7. If $f(x)=x+4$ and $h(x)=4x-1$. Find a function $g$ such that $g\circ f=h$.

8. Find the exponential function $f(x) = Ca^x$ whose graph is given.



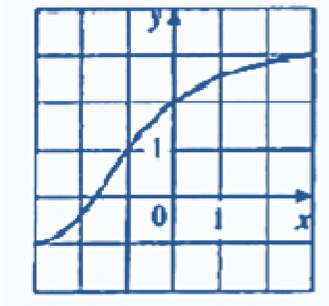

9. The graph of $f$ is given.

(a) Why is f one-to-one?

(b) What are the domain and range of $f^{-1}$?

(c) What is the value of $f^{-1}(2)$?

(d) Estimate the value of $f^{-1}(0)$.




10. Find a formula for the inverse of the function.

$$f(x) = \frac{4x-1}{2x+3}$$

11. Find the exact value of each expression.

(a) $e^{-2\ln 5}$ (b) $\ln\left(\ln e^{e^{10}}\right)$

12. Make a rough sketch of the graph of each function. Do not use a calculator. Start with the basic graph, apply the transformation if necessary.

(a) $y = \ln(-x)$ (b) $y = \ln|x|$

13. Solve each equation for $x$.

(a) $e^{7-4x} = 6$ (c) $\ln(\ln x) = 1$

(b) $\ln(3x-10) = 2$ (d) $e^{ax} = Ce^{bx}, a \neq b$

14. Given $f(x) = \ln(2 + \ln x)$, find

(a) The domain of $f$.

(b) $f^{-1}$ and its domain.

15. Given the function $x = 1 + 3t,\ y = 2 - t^2$,

(a) Sketch the curve by using the parametric equations to plot points. Indicate with an arrow the direction in which the curve is traced as $t$ increases.

(b) Eliminate the parameter to find a Cartesian equation of the curve.

16. Describe the motion of a particle with position $(x, y)$ as $t$ varies in the given interval.

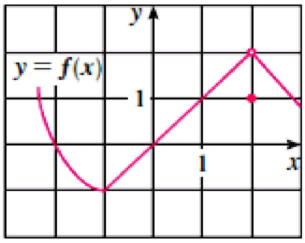
$$x = 3 + 2\cos t,\ y = 1 + 2\sin t,\ \frac{\pi}{2} \le t \le \frac{3\pi}{2}.$$

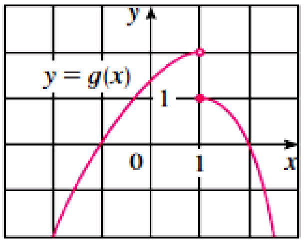
T2: Limits and Derivatives

1. Sketch the graph of the function and use it to determine the values of a place for which $\lim_{x\to a} f(x)$ exists.

$$f(x)=\begin{cases} 1+x & if \quad x<1 \\ x^2 & if \quad -1\le x<1 \\ 2-x & if \quad x\ge 1 \end{cases}$$

2. The graph of $f$ and $g$ are given. Use them to evaluate each limit, if it exists. If the limit does not exist, explain why.




(a) $\lim\limits_{x\to 2}[f(x)+g(x)]$

(b) $\lim\limits_{x\to 1}[f(x)+g(x)]$

(c) $\lim\limits_{x\to 0}[f(x)g(x)]$

(d) $\lim\limits_{x\to -1}\dfrac{f(x)}{g(x)}$

(e) $\lim\limits_{x\to 2}x^3f(x)$

(f) $\lim\limits_{x\to 1}\sqrt{3+f(x)}$

3. Evaluate the limit, if it exists.

(a) $\lim\limits_{x\to -1}\dfrac{2x^2+3x+1}{x^2-2x-3}$

(b) $\lim\limits_{h\to 0}\dfrac{\sqrt{1+h}-1}{h}$

(c) $\lim\limits_{x\to -4}\dfrac{\sqrt{x^2+9}-5}{x+4}$

(d) $\lim\limits_{t\to 0}\left(\dfrac{1}{t}-\dfrac{1}{t^2+t}\right)$

4. Use the Squeeze Theorem to show that

$$\lim_{x\to 0}\sqrt{x^3+x^2}\sin\frac{\pi}{x}=0$$

Illustrate by graphing the functions $f$, $g$ and $h$ (in the notation of the Squeeze Theorem) on the same screen.

5. If $2x\le g(x)\le x^4-x^2+2$ for all $x$, evaluate $\lim\limits_{x\to 1}g(x)$.

6. Prove that $\lim\limits_{x\to 0}x^4\cos\dfrac{2}{x}=0$.

7. Find the limit, if it exists. If the limit does not exist, explain why.

$$\lim_{x\to -6}\frac{2x+12}{|x+6|}$$

8. Use the definition of continuity and the properties of limits to show that the function is continuous at the given number $a$.

$$h(t)=\frac{2t-3t^2}{1+t^3},\ a=1$$

9. Use the definition of continuity and the properties of limits to show that the function $g(x)=2\sqrt{3-x}$ is continuous on the interval $(-\infty,3]$.

10. Use the Intermediate Value Theorem to show that there is a root of the given equation in the specified interval.

$$x^4+x-3=0,\quad (1,2).$$

11. Find the limit.

(a) $\lim\limits_{x\to 2^+} e^{3/(2-x)}$ (b) $\lim\limits_{x\to 3^+} \ln\left(x^2-9\right)$ (c) $\lim\limits_{x\to\infty}\dfrac{\sin^2 x}{x^2}$

12. (a) Find the slope of the tangent line to the parabola $y=4x-x^2$ at the point (1, 3)

(i) using $\lim\limits_{x\to a}\dfrac{f(x)-f(a)}{x-a}$ (ii) using $\lim\limits_{h\to 0}\dfrac{f(a+h)-f(a)}{h}$

(b) Fine an equation of the tangent line part (a).

13. (a) Find the slope of the tangent to the curve $y=1/\sqrt{x}$ at the point where $x=a.$

(b) Find equations of the tangent lines at the points $(1,1)$ .

T3: Differentiation Rules

1. Differentiate the function.

(a) $y = \sqrt{x}(x-1)$ (b) $y = ae^v + \frac{b}{v} + \frac{c}{v^2}$

2. Differentiate:

(a) $F(y) = \left(\frac{1}{y^2} - \frac{3}{y^4}\right)(y + 5y^3)$ (b) $y = \frac{1}{s + ke^s}$ (c) $f(x) = \frac{x}{x + \frac{c}{x}}$

3. If $g(x) = x/e^x$, find $g^{(n)}(x)$.

4. Differentiate:

(a) $y = \sec\theta\tan\theta$ (b) $y = \frac{1 - \sec x}{\tan x}$

5. Find the derivative of the function.

(a) $y = \cos(a^3 + x^3)$ (b) $y = e^{x\cos x}$ (c) $y = \frac{r}{\sqrt{r^2 + 1}}$ (d) $y = 2^{\sin \pi x}$

6. Find $dy/dx$ by implicit differentiation.

(a) $2\sqrt{x} + \sqrt{y} = 3$ (b) $4\cos x \sin y = 1$

7. Differentiation the function.

(a) $f(x) = \sqrt[5]{\ln x}$ (b) $f(x) = \log_2(1 - 3x)$ (c) $F(t) = \ln\frac{(2t+1)^3}{(3t-1)^4}$

8. Use logarithmic differentiation to find the derivative of the function.

(a) $y = x^x$ (b) $y = (\sqrt{x})^x$

9. Find the linear approximation of the function $f(x) = \sqrt{1-x}$ at $a = 0$ and use it to approximate the number $\sqrt{0.9}$ and $\sqrt{0.99}$.

10. In a precision manufacturing process, ball bearings must be made with a radius of 0.6 mm, with a maximum error in the radius of ±0.015 mm. Estimate the maximum error in the volume of the ball bearing.

T4: Applications of Differentiation

1. If a snowball melts so that its surface area decreases at a rate of 1cm$^2$/min, find the rate at which the diameter decreases when the diameter is 10cm.

2. A plane is flying directly away from you at 500 mph at an altitude of 3 miles. How fast is the plane's distance from you increasing at the moment when the plane is flying over a point on the ground 4 miles from you?

3. Find the absolute maximum and absolute minimum value of f on the given interval.

   (a) $f(x)=\frac{x^2-4}{x^2+4}, [-4,4]$ (b) $f(t)=2\cos t+\sin 2t, [0,\pi/2]$

4. (a) Find the intervals on which *f* is increasing or decreasing.
   (b) Find the local maximum and minimum values of *f*.
   (c) Find the intervals of concavity and the inflection points.

   (i) $f(x)=2x^3+3x^2-36x$ (ii) $f(x)=e^{2x}+e^{-x}$

5. Find the limit. Use L'Hopital Rule where appropriate. If there is a more elementary method, consider using it. If L'Hopital Rule doesn't apply, explain why.

   (a) $\lim_{t\to 0}\frac{e^t-1}{t^3}$ (b) $\lim_{\theta\to \pi/2}\frac{1-\sin\theta}{\csc\theta}$ (c) $\lim_{x\to 0^+}\frac{\ln x}{x}$ (d) $\lim_{x\to\infty}\frac{(\ln x)^2}{x}$

   (e) $\lim_{x\to\infty}\frac{e^{u/10}}{u^3}$ (f) $\lim_{x\to 1}\left(\frac{x}{x-1}-\frac{1}{\ln x}\right)$

6. Consider the following problem: A farmer with 750ft of fencing wants to enclose a rectangular area and the divide it into four pens with fencing parallel to one side of the rectangle. What is the largest possible total area of the four pens?

7. A rectangular storage container with an open top is to have a volume of 10m$^3$. The length of its base is twice the width. Material for the base costs \$10 per square meter. Material for the sides cost \$6 per square meter. Find the cost of materials for the cheapest such container.

8. Use Newton's method to approximate the given number correct to eight decimal places.

$$\sqrt[5]{20}$$

9. Use Newton's method to find the two roots of equation correct to eight decimal places. Use –1.95 and –0.8 as the initial approximations respectively.

$$x^2\sqrt{2-x-x^2}=1$$

T5: Integrals

1. Evaluate the integral.

   (a) $\int_{1/2}^{\sqrt{3}/2} \frac{6}{\sqrt{1-t^2}}\, dt$  (b) $\int_{-1}^{1} e^{u+1}\, du$  (c) $\int \cos^3\theta \sin\theta \, d\theta$

2. Evaluate the indefinite integral.

   (a) $\int \frac{a+bx^2}{\sqrt{3ax+bx^3}}\, dx$  (b) $\int e^x \sqrt{1+e^x}\, dx$  (c) $\int \frac{\sin 2x}{1+\cos^2 x}\, dx$

3. Evaluate the definite integral $\int_e^{e^4} \frac{dx}{x\sqrt{\ln x}}$

4. Evaluate the integral.

   (a) $\int \sqrt{x} \ln x \, dx$  (b) $\int_1^2 (\ln x)^2\, dx$  (c) $\int \sec^3 x \, dx$

5. Evaluate the integral.

   (a) $\int \frac{2x^2+5}{(x^2+1)(x^2+4)}\, dx$  (b) $\int \frac{x^3+4}{x^2+4}\, dx$

6. If $f(x) = \int_0^{\sin x} \sqrt{1+t^2}\, dt$ and $g(y) = \int_3^y f(x)\, dx$, find $g''(y)$ and hence $g''(\pi/6)$.

7. Use (a) the Trapezoidal Rule (b) Simpson's Rule to approximate the given integral with the specified value of *n*. Round your answers to six decimal places.

$$\int_0^4 \sqrt{1+\sqrt{x}}\, dx, \; n=8$$

8. (a) Find the approximations $T_{10}$ and $S_{10}$ for $\int_0^{\pi} \sin x \, dx$ and corresponding errors $E_T$ and $E_s$.

   (b) How large do we have to choose *n* so that the approximations $T_n$ and $S_n$ to the integral in part (a) are accurate to within 0.00001= $10^{-5}$ ?